

BUas Data Management Plan (DMP)	
0. General information	
Project:	Project Name: SAM Automation - Predictive Maintenance for Industrial Robots Project Institution: Breda University of Applied Sciences (BUas) Project Project Duration: September 2024 - January 2025 (18 weeks)
Team:	Chrislande Duterloo, Monieka Hardjosoedarmo, Viktória Kubišová, Bartosz Kudyba, Kajetan Noweś
Name of RDM support staff: (Funders, such as NWO, expect you to consult RDM support staff to complete the DMP)	Name: Shival Indermun (Lecturer), Ph.D Position: Project Mentor BUas Email: indermun.s@buas.nl Name: Mathias Van Quathem Position: Ethics Supervisor BUas Email: quathem.m@buas.nl
Date of consulting RDM support staff:	Date of consultation with support staff (Shival): 10/12/2025
1. Collecting data	
In this project, newly to be collected data/existing data(bases) are used:	☒ Yes, newly to be collected data are used ☐ Yes, Existing data are used ☐ No This project will generate entirely new data from controlled laboratory experiments at BUas. No existing datasets will be re-used because the MVP requires: 1. Real-time sensor data from specific robots (Niryo Ned2, UFactory 850) with custom external sensors (pressure, accelerometer) and Arduino-based preprocessing 2. Two baseline "healthy loop" datasets (basic block stacking and advanced conveyor belt integration) collected over 4-6 weeks to establish normal operational parameters 3. Controlled fault scenario simulations using custom 3D-printed objects to replicate real factory technical failures (vacuum leaks, clogged filters, misalignment, overloaded payloads, rough surfaces) 4. Synchronized multi-modal data streams (robot telemetry at 2.5 Hz, external sensors at 10 Hz) with ground-truth fault labels Existing public datasets do not provide the controlled fault scenarios, multi-modal sensor integration, or specific robot configurations needed for this predictive maintenance project. All data will be collected fresh during the project timeline to ensure consistency, reproducibility, and alignment with the specific solution objectives.
The collected data are suitable/unsuitable for reuse (see also under 4). (If the data is not suitable or not meant for reuse, it should be explained why)	☐ Yes ☒ No, because: The collected data is not suitable for reuse beyond this specific MVP demonstration due to: • Data collected from specific educational robot models (Niryo Ned2, UFactory 850) in controlled lab environment that does not represent real factory conditions

that is the case. Data can, for example be anonymised or summarised. Metadata is delivered along with the research data, and needs to remain available at any rate, and needs to be disclosed.)	• Custom sensor integration and Arduino preprocessing specific to this MVP setup • Fault scenarios simulated using project-specific 3D-printed objects and lab-based methods (hairdryer for heating, manual valve adjustments) that do not replicate real equipment degradation • Baseline patterns specific to small-scale educational robots handling gram-weight plastic blocks, fundamentally different from industrial robots handling kilogram-scale production items Purpose of data: Demonstrate proof-of-concept that the pipeline architecture can detect anomalies and generate alerts. The data is not intended for production deployment or retraining on factory systems. Real factory implementation would require: Entirely new data collection on actual production robots, in real factory environments, handling actual production items, with industrial-grade sensors. Each factory deployment needs site-specific baseline data (4-6 weeks minimum) and complete model retraining. What is reusable: The pipeline architecture, methodology, and documentation approach—not the actual collected data. While the data technically suitable for reuse due to comprehensive documentation (codebook with 92 variables), open standard formats (Parquet, SQLite), clear provenance (who, when, how collected), and adherence to FAIR principles, it serves as reference for MVP development only and has no practical reuse value for other projects or production systems.
State the nature and content of data that is collected:	In this project, the following data is collected: Quantitative numeric time-series data in tabular format and textual documentation. TIME-SERIES SENSOR DATA (Primary Dataset) Kind: Numeric time-series quantitative data Format: Apache Parquet (.parquet) - open, columnar storage format Structure: 92 columns per record including: Robot Telemetry (collected via SSH/ROS at ~2.5 Hz): • Joint positions: j1, j2, j3, j4, j5, j6 (radians, float64) • Motor temperatures: motor1_temp through motor6_temp (°C, int64) • Raspberry Pi temperature: rpi_temp (°C, int64) • Motor voltages: motor1_voltage through motor6_voltage (V, float64) • Derived metrics: avg_motor_temp, max_motor_temp, avg_motor_voltage (float64) • Error state: error_state (binary flag, int64) External Sensor Data (collected via Arduino/UDP at ~10 Hz, synchronized): • Vacuum pressure: pressure (raw ADC value, int64) • Vibration statistics: vib_mean, vib_max, vib_min, dominant_freq (float64) • FFT frequency bins: 64 vibration frequency bands from 1.6 Hz to 98.4 Hz (vib_1.6, vib_3.1, ..., vib_98.4) preprocessed by Arduino (float64) Timestamps: UTC timestamp (datetime64[ns]) with ~400ms average interval

	Type of files: • Apache Parquet (.parquet) - time-series sensor data (open, columnar storage format) Format Rationale: Parquet provides efficient compression (60-70% size reduction vs. CSV), fast columnar read/write operations, preserves data types, and is compatible with standard scientific Python tools (pandas, arrow). It is listed as a preferred format by 4TU.ResearchData and DANS for long-term preservation and data sharing. Size of files: ☒ 0-10 GB ☐ 10-100 GB ☐ 100-1000 GB ☐ > 1000 GB
Metadata	In this project, the following metadata is collected: Session-Level Metadata: Plain text files (experiment_meta_YYYYMMDD_HHMMSS.txt) documenting session information (date, duration, robot IP), experiment configuration (blocks per cycle, conveyor settings, z-offsets), and results (cycles completed, blocks moved, exit reason). Variable-Level Metadata: Codebook documenting all 92 variables with names, descriptions, data types, units, valid ranges, and sampling rates. Experimental Context: README file with consolidated documentation describing experimental conditions, fault simulation methods, and observational notes across all data collection sessions
2. Storage of data and metadata	
State how data storage is organized, described and documented.	Data storage is organized using a timestamped folder structure with standardized file naming conventions. Each data collection session is stored in a separate folder named YYYYMMDD_HHMMSS containing three Parquet files: robot telemetry only, external sensor data only, and the merged unified dataset. This three-file structure preserves data provenance and enables debugging and verification. Metadata is stored as plain text files (experiment_meta_YYYYMMDD_HHMMSS.txt) in a dedicated metadata subfolder, documenting session information, experiment configuration, and results. A comprehensive README.md file in the root data folder provides consolidated documentation of experimental conditions, fault simulation methods, and observational notes across all sessions. Additional project documentation is stored in the documentation/ folder at the project root and includes: • Codebook: Comprehensive documentation of all 92 variables with descriptions, data types, units, valid ranges, and sampling rates • Technical Design Document: System architecture, sensor integration, data flow, hardware specifications, dashboard design, backend infrastructure, modeling approaches, data cleaning and preprocessing procedures

	• Data Management Plan: This document outlining data collection, storage, retention, and sharing protocols • FAIR Data Checklist: Completed assessment of Findable, Accessible, Interoperable, and Reusable principles • FAIR Data Explanation: Summary document explaining how the data meets FAIR principles • GDPR Compliance Checklist: Assessment documenting non-applicability of GDPR to sensor-only data • Ethics & Legal Framework Assessment: Comprehensive evaluation of applicable frameworks (GDPR, FAIR, ALTAI, DPIA, EU AI Act, DEDA) and responsible AI documentation • Data Storage Protocol: Technical specifications for file organization, backup procedures, and access control During the project, this facility can only be accessed by researchers designated by the project leader/researcher, being: • Chrislande Duterloo • Monieka Hardjosoedarmo • Viktória Kubišová • Bartosz Kudyba • Kajetan Noweś • BUas Teacher Team (advisor access) • Stan Jacobs (access on request)
Folder structure	<i>Note: This structure shows only the data and documentation folders referenced in the above subsection. The complete project repository contains additional folders (src, dashboard, nyrio_scripts, etc.) not listed here.</i> 2025-26ab-fai3-specialisation-project-team-sam_automation • data/ (root data folder) ○ README.md (consolidated session descriptions and experimental documentation) ○ metadata/ (session metadata files) ▪ experiment_meta_YYYYMMDD_HHMMSS.txt ○ YYYYMMDD_HHMMSS/ (timestamped session folders - raw data) ▪ robot_data_robot_[timestamp].parquet (raw robot telemetry) ▪ robot_data_sensor_[timestamp].parquet (raw external sensor data) ▪ robot_data_unified_[timestamp].parquet (processed/merged dataset) • documentation/ ○ SAM_Automation_Codebook.md ○ SAM_Automation_Technical_Design_Document.pdf ○ SAM_Automation_Data_Management_Plan.pdf ○ SAM_Automation_FAIR_Checklist.pdf ○ SAM_Automation_FAIR_Summary.pdf ○ SAM_Automation_GDPR_Checklist.pdf ○ SAM_Automation_Framework_Assessment.md ○ SAM_Automation_Data_Storage_Protocol.pdf
Physical data	Not applicable. This project does not generate or collect physical data. All data is digital (time-series sensor data, metadata files, and documentation). And kept for 2 years until student team graduation in 2027.

Risks	The research entails the following risks with regard to security, damage and/or loss of data: All data is stored in the project's GitHub repository accessible to the team and supervisors. Primary risks include accidental deletion or overwriting of files through Git operations, repository corruption, and reliance on GitHub's infrastructure availability. Local laptop storage during active data collection creates temporary single point of failure until files are committed and pushed to the repository. Damage and loss of data are avoided by means of: Data loss is mitigated through Git version control, which maintains complete history of all changes and enables recovery of previous versions. The distributed nature of Git means each team member has a full copy of the repository on their local machines, providing multiple backup points. Supervisors access the repository online via GitHub. The timestamped folder structure and three-file architecture (separate raw robot, raw sensor, and merged files) prevent accidental overwrites. GitHub's infrastructure provides reliable cloud storage with automated backups. Access control is managed through GitHub repository permissions, restricting access to project team members and BUAs supervisors only. The Parquet format includes built-in data integrity checks to detect file corruption. Regular commits and pushes after each data collection session ensure minimal data loss risk, with the maximum potential loss limited to uncommitted work from the current session. Upon project completion, the complete repository will be shared with SAM Automation as project deliverables for review and reference. The data is expected to be preserved by BUAs until student team graduation in 2027, after which preservation follows standard BUAs institutional repository policies.
Data of partners	The results of partners outside BUAs participating in the project are stored at BUAs and the relevant partners, in the following manner: Not applicable. The external partner (SAM Automation) does not provide or collect data for this project. All data is collected by the BUAs student team using BUAs lab equipment. Upon project completion, the final datasets and documentation will be shared with SAM Automation as deliverables, but SAM does not generate or store data during the research phase.

3. Personal data

When research is conducted, personal data are collected, processed and stored	☐ Yes; Observing legislation on personal data is guaranteed at BUAs by the Executive Board and the Academy Director, respectively, all this according to the mandate in the rules on jurisdiction, supervised by the Data Protection Officer ☒ No
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4. Accessibility, Archiving and Publishing research data.

State the place where data are archived, digitally and/or physically, how data is secured and for whom data is accessible/usable.	The datasets are stored in the project's GitHub repository (2025-26ab-fai3-specialisation-project-team-sam_automation), accessible via standard Git version control protocols. Data is secured through GitHub repository permissions restricting access to authorized project team members and BUAs supervisors. No encryption is applied to the data files as they contain only technical robot sensor measurements without any sensitive or personal information. The data will be preserved according to BUAs institutional policies for student project repositories. The GitHub repository remains accessible to the student team during their studies (until 2027) and is shared with SAM Automation upon project completion (January 2026) as project deliverables. Long-term preservation beyond the student period follows standard BUAs data management protocols.
Back-ups	Backups are managed automatically through Git version control. Each team member's local repository clone serves as a distributed backup, and GitHub provides cloud-based storage with automated backups. Commits and pushes occur after each data collection session (typically multiple times per week during active data collection periods), ensuring all data is synchronized to the remote repository.
Destruction of data	The following data must be destroyed because of contractual obligations/legislation: Not applicable. The project does not collect personal data, sensitive information, or any data subject to mandatory destruction requirements under GDPR or other legislation. All collected data consists of technical robot sensor measurements without confidentiality restrictions. The decision to store/destroy data will be taken by: Data preservation and destruction decisions follow BUAs institutional policies for student project repositories. Upon project completion, the student team and BUAs supervisors determine the retention period in accordance with institutional data management protocols. SAM Automation may retain their copy of project deliverables according to their internal data management policies.
(Limited) Accessibility	The following data will be accessible to a limited extent: All project data (time-series sensor data, metadata, documentation) is accessible only to authorized project team members, BUAs supervisors (via GitHub repository permissions), and SAM Automation (upon project completion as project deliverables). Limited accessibility and/or disclosure is based on: The data is project-specific for MVP demonstration purposes only and has no practical reuse value beyond this project scope. Limited access ensures proper project management and prevents unauthorized distribution of data that is specific to the BUAs lab setup and SAM Automation collaboration.

	The following data will not be disclosed: The complete dataset will not be made publicly available. Data remains restricted to the project team, BUas supervisors, and SAM Automation (as project deliverables upon completion). The following data will be available for reuse: None. As documented in Section 1, the collected data is not suitable for reuse due to being specific to educational robot models in controlled lab environments that do not represent real factory conditions. The data serves as proof-of-concept for the pipeline architecture only and has no practical reuse value for other projects or production systems. State the person(s) who will have access, and what actions these persons will be authorized to take (only reading/processing/deleting) • Project team members (5 students): Full access (reading, processing, committing, pushing, deleting) via GitHub repository • BUas supervisors: Read access via GitHub repository for monitoring and assessment purposes • SAM Automation (post-project): Read access to complete repository deliverables for reference and review purposes Access to the data can be granted by: GitHub repository access is managed by the project team lead in coordination with BUas supervisors. New access requests are evaluated based on legitimate project-related need.
Data Archiving and publishing	The following data are available for disclosure and publication: None. The data will not be publicly disclosed or published as it is not suitable for reuse beyond this specific project (see Section 1 for detailed justification). The data for reuse will be stored and made available in: Data repository: Not applicable - data not deposited in public repository. A persistent identifier will be made use of. ☐ Yes ☒ No Not applicable - no public repository deposit, therefore no DOI or persistent identifier assigned. Data will be reused in the context of the licence (e.g. Creative Commons): Not applicable. Internal project data with restricted access (project team, BUas supervisors, SAM Automation only). No public license applied. The data repository used is certified: ☐ Yes ☐ No

	Not applicable - no public repository used. If the repository is not certified, the following minimum criteria will hold: Not applicable - data stored in private GitHub repository for internal project use only, not in a public data repository.
Reusability	To gain access to the data and reuse this data, users need the following tools and/or software: While the data is not suitable for practical reuse (as documented in Section 1), technical access to read and process the data requires: For data files: • Python 3.11.13 with libraries specified in the requirements.txt file in the repository (including but not limited to: pandas, fastparquet, matplotlib, streamlit). • Text editor for reading plain text metadata files (.txt) and README documentation (.md) For repository access: • Git version control system • GitHub account with repository permissions Minimum system requirements: • Any operating system supporting Python (Windows, macOS, Linux) • Sufficient RAM to load datasets (~10 GB storage space, ~4 GB RAM recommended for data processing) Documentation: All variable definitions, data formats, file organization, and processing procedures are documented in the Codebook, Technical Design Document, and Data Storage Protocol included in the repository's documentation folder.
Metadata and Current Research Information System	Besides registering the metadata in the data repository (Zenodo or other repository?): Not applicable. The data is not deposited in a public data repository. The metadata will be registered in the Current Research Information System of BUAs; Pure: ☐ Yes ☒ No The session-level metadata consists of basic technical parameters (date, duration, conveyor settings, cycles completed) specific to individual data collection runs. This metadata is stored as plain text files within the project repository for internal reference and does not warrant registration in external systems. Complete project documentation (codebook, technical design document, data management plan, FAIR and GDPR checklists...) is included in the repository's documentation folder.

5. Costs

(Extra) costs for research data management	(Extra) costs for research data management ☒ No special resources or facilities will be required for the project Justification:
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	All data management is conducted using existing BUas infrastructure and freely available tools: • Storage: GitHub repository (free educational account), local laptop storage (team members' existing equipment) • Software: Open-source tools (Python, Git, pandas, pyarrow) with no licensing costs • Hardware: BUas lab equipment (Niryo Ned2, UFactory 850 robots, sensors) already available • Computational resources: Standard laptops sufficient for data processing (~10 GB storage, 4 GB RAM) • Personnel: Student team conducting all data collection, processing, and documentation as part of project work • Physical facilities: BUas robotics lab accessed during regular project hours No additional financial resources, specialized IT infrastructure, external storage services, or extra staff time beyond normal project allocation are required.
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6. Evaluation and embedding

The research plan and the DMP must be evaluated at regular intervals, for example with the supervisor/project leader, a research group, an external partner.	The Data Management Plan is evaluated during milestone presentations with the BUas mentor and SAM Automation client throughout the project (October 2024 - January 2025) and when significant changes to data collection, storage, or project scope occur. The DMP has undergone review by the BUas ethics supervisor for compliance with institutional ethical and legal requirements.
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7. Appendices

The following appendices are part of this DMP:	• Codebook (variable documentation for 92 data columns) • Technical Design Document (system architecture, sensor integration, data flow, hardware specifications, dashboard design, backend infrastructure, modeling approaches, data cleaning and preprocessing) • Ethics & Legal Framework Assessment (comprehensive evaluation of GDPR, FAIR, ALTAI, DPIA, EU AI Act, DEDA frameworks and responsible AI documentation, reviewed by BUas ethics supervisor) • FAIR Data Checklist (completed assessment of Findable, Accessible, Interoperable, Reusable principles)
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	• FAIR Data Explanation (summary of how data meets FAIR principles) • GDPR Compliance Checklist (assessment documenting non-applicability to sensor-only data) • Data Storage Protocol (file organization, backup procedures, access control specifications) • Business Requirements Document (BRD) • README files (data folder documentation)
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